



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

🌐 Website: www.ktunotes.in

Altknater

$N_s = \frac{120f}{P}$ → generator → runs → Syn speed.

Basic principle.

- Electromagnetic Induction
- when ever a conductor cuts mag. field, emf is induced in it
- $e \propto \frac{d\phi}{dt}$
- consist of ϕ Arm. winding & field winding.
- Arm → stator field winding → Rotor
- field winding is energised → DC supply & rotated by a prime mover
- Thus Arm. conductors cuts the mag. field of field winding.
- Thus by Elec. Indu. principle, emf is induced in stator / Arm. winding.
- induced emf is alternating

Construction

Stator,

Rotor,

- salient pole
- non-salient pole

- rotor poles are projected
- more no. of poles.
- large dia & small axial.
- low & medium speed.
- hydraulic turbine
- const. difficult
- air gap present - flux is not uniformly distributed
- less sinusoidal wave.

Advantages of stationary Arm

1. Easy to insulate
2. vol. sliding ↑
3. slip current - collect directly
3. can withstand mech stress & centrifugal force.
4. Rotor - weight ↓ can run at high speed.
5. only 2 slip-rings used
6. sparking ↓
7. mech. balance.
8. heavy bearing is not required.

Types of winding

1. single layer

- One slot contains only one coil sides of a coil.
- closed or semi-closed slots.
- insulation is easy.

2. Double layer

- 2 coil sides of different coils is placed inside a slot. → Top & bottom.
- open slot.

Advantages

- End connections become simple.
- Economical to manufacture
- provide neat arrangements
- coil span can be easily selected.

3. Full pitched.

- Coil span = pole pitch coil span < pole pitch
- 180° $180 - \alpha$

- E_{C1} & E_{C2} are in phase with each other
- are not in phase
- E_{C2} is displaced by ' α '

$$E_C = 2E$$

Pitch factor, k_c .

$$k_c = \frac{\text{emf induced - short pitch}}{\text{emf induced - full pitch}}$$

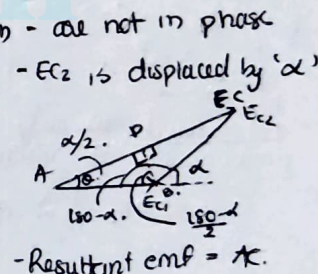
$$k_c = \frac{AC}{2E} = \frac{2E \cos \frac{\alpha}{2}}{2E}$$

$$k_c = \cos \frac{\alpha}{2}$$

Advantages of short pitch

1. Savings of cu.
2. ↓ harmonics
3. smooth sinusoidal wave form current
5. Efficiency is ↑

4. short pitched.



$$\Delta AOB$$

$$180 = 90 + \theta + \frac{180}{2}$$

$$\alpha = 2\theta$$

$$\theta = \frac{\alpha}{2}$$

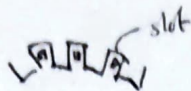
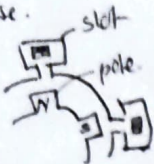
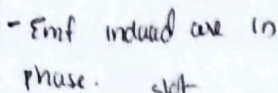
$$AO = AB \cos \frac{\alpha}{2}$$

$$AC = AO + OC$$

$$= 2AO$$

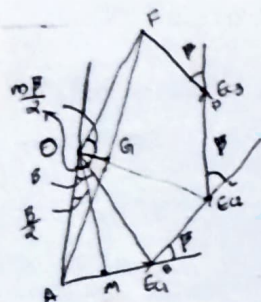
Concentrated

- No. of poles = slots = no. of coil sides.
- slot / pole = 1
- one coil side is inside one slot under one pole



Advantages

1. sinusoidal wave form
2. ↓ harmonics
3. ↓ amp. gradient
4. cooling is efficient



For distributed winding
phasor sum of voltages = Af

$$= 2AG = 2OA \sin \left[n \frac{\beta}{2} \right]$$

For concentrated winding

$$E_c = AB + BC + CD + \dots \quad (m \text{ Nos})$$

$$= m \cdot AB.$$

$$= m \cdot AB \quad \xrightarrow{\quad} \quad AB = 2AM$$

$$= m \cdot 0.5 \sin\left(\frac{B}{2}\right) \quad = 20.6 \sin\left(\frac{B}{2}\right)$$

Kcd = Emf $\rightarrow$ Distributed code

Emf $\rightarrow$ concentrated wdg.

$$= \frac{\cancel{20A} \sin\left(m\frac{B}{2}\right)}{m\cancel{20A} \sin\left(\frac{B}{2}\right)} = \frac{\sin\left(m\frac{B}{2}\right)}{m \cdot \sin\left(\frac{B}{2}\right)}$$

6. Distribuko

$$\left. \begin{array}{l} \text{Avg value of gen} \\ \text{emf per convy} \end{array} \right\} = \frac{\text{Peak val in one revolution}}{\text{Time for one revolution in sec}}$$

$$E_{avg} = \frac{d\phi}{dt}$$

$$E_{\text{avg per cond}} = \frac{\phi \cdot P}{\frac{60}{N}} = \frac{\phi P N}{60} = \frac{\phi P 120 f}{60 P}$$

$$= 2 \phi \cdot f$$

$$E_{avg} \text{ per phase} = 2 \phi f \cdot 2 = 4 \phi f \cdot T.$$

$$E_{rms} \text{ per phase} = 4.44 \phi f T.$$

For short pitched & distributed wdg =

4.44. ϕ . f. T. kd. kc.

Harmonics in gene. emb.

- cause of harmonics is the non-sinusoidal field flux wave form in air gap.
- Space harmonics - Slot harmonics

Space harmonics

- non-sinusoidal distribution of the coils in the machine & slotting.
- ie, machine air gap mmf & flux are non-sinusoidally distributed in space
- order of space harm. = $6k \pm 1$,

Slot harmonizes

- caused due to variable reluctance caused by slotting the stator & rotor.
- order of harmonics = $\frac{S}{P} \pm 1$

eliminated by

- b. Air gap - small at pole center
- large at pole end.

2. Skewing salient poles

slots

4. 3rd harmonics eliminated by connecting arm.

winding in Y or Δ

5. By using short pitch winding

$$k_{cn} = 0, \cos \frac{n\alpha}{2} = 0 \quad \frac{n\alpha}{2} = 90^\circ$$

$$\alpha = 60^\circ$$

By making $\alpha = 60^\circ$ we can eliminate 3rd harmonics

6. By using distributed winding

$$k_{dn} = 0, \frac{\sin n \frac{m\beta}{2}}{n \sin \frac{\beta}{2}} = 0$$

$$\sin 5 \frac{m\beta}{2} = 0; \quad 5 \frac{m\beta}{2} = 180^\circ$$

$$m\beta = 72^\circ;$$

By making $m\beta = 72^\circ$ we can eliminate 5th harmonics

Armature Reaction

- The effect of armature flux on the field flux is called armature reaction.
- The armature flux will distort, oppose or aid the flux depending on the load p.f

UPF

- Armature current & induced emf - same phase.
- Emf induced in the arm. due to the changing main field flux linked with arma.

- main field flux

$$\phi_f = \phi_{fm} \sin \omega t$$

- Emf across armature

$$\frac{d\phi_f}{dt} = -\omega \phi_{fm} \cos \omega t$$

- Arma. flux ϕ_a is in phase with armature current I_a .

- At upf I & E are same phase.

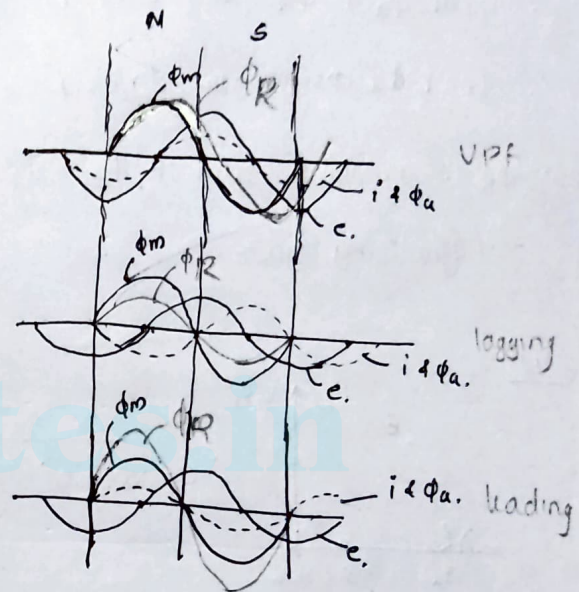
$\therefore \phi_a$ is phase with E

- ϕ_f is quadrature with E

$\therefore \phi_a$ is quadrature with ϕ_f

- two fluxes are $\perp$ to each other

$\therefore$ AR at upf is purely distorting or cross magnetizing type.



Lagging p.f

$$\phi_{fm} = \phi_{fm} \sin \omega t$$

$$E \propto -\frac{d\phi_f}{dt}$$

$$E \propto -\omega \phi_{fm} \cos \omega t$$

- ϕ_m got max. value 90° before E

- ϕ_m leads E by 90°

- $I_a \propto \phi_a$

- I_a lags E by 90°

$\therefore \phi_a$ lags E by 90°

$\therefore \phi_a$ & ϕ_f act directly opposite to each other.

- AR is purely demagnetizing type.

- Φ_a weakens Φ_m .

Leading P.F

- Φ_m leads E by 90°

- $I_a \propto \Phi_a$.

- I_a leads E by 90°

- $\therefore \Phi_a$ leads E by 90°

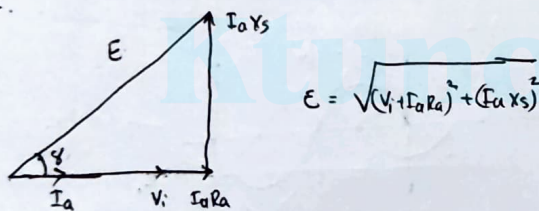
- $\therefore$ both Φ_a & Φ_m lead E by 90°

$\therefore \Phi_f$ & Φ_a are in same direction

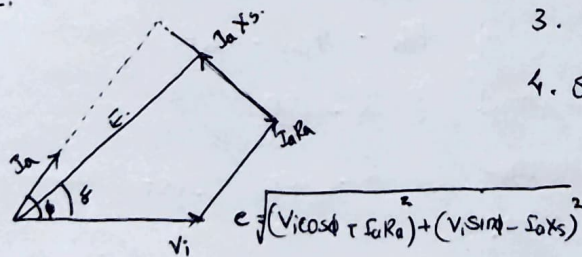
$\therefore \Phi_R$ is arithmetic sum of Φ_a & Φ_f

- magnetizing type

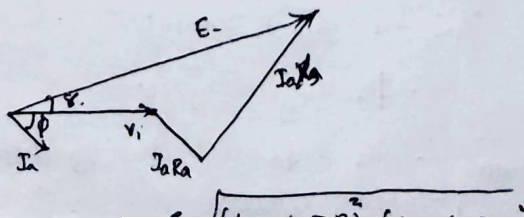
UPF



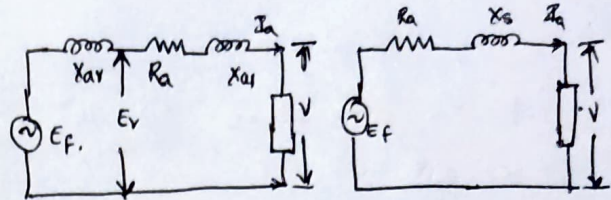
Leading P.F.



Lagging P.F.



Equivalent ckt



Change in terminal vol.

- R_a - arm. Resistance.

- X_{ar} - arm reaction

- X_{al} - arm leakage reactance

$$X_s = X_{ar} + X_{al}$$

$$Z_s = R_a + jX_s$$

$$V = E_f - I_a R_a - j I_a (X_{ar} + X_{al})$$

E_r - air gap emf ; Φ_r - air gap flux.

Φ_f - main flux ; Φ_{ar} - arma. flux.

$$\Phi_r = \Phi_f + \Phi_{ar}$$

Θ - Syn. impedance angle $\tan^{-1} \frac{X_s}{R_a}$.

Losses and Efficiency

1. No load rotational losses (fric, wind, oc loss)
2. field ckt loss - $I^2 R$ - field winding
3. Direct load loss - $I^2 R$ - arm winding
4. Stray loss -

1. Blondel's Two Reaction theory.

- Non-salient pole $\rightarrow$  $\rightarrow$ air gap is uniform
- flux is uniformly distributed along the pole surface.
- In the case of non-salient pole  x_d and x_q

$\rightarrow$ Two axes are recognized in salient pole machines, namely on axis along the polar axis called direct axis and axis along the inter-polar axis called quadrature axis.

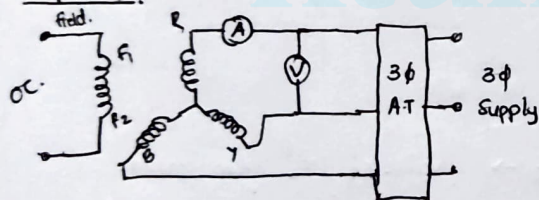
I_a resolved into

- $\rightarrow I_d \perp$ to E_o
- $\rightarrow I_q$ along E_o .

$x_d \rightarrow$ direct axis syn react.

$x_q \rightarrow$ quadrature axis syn react.

Slip Test.

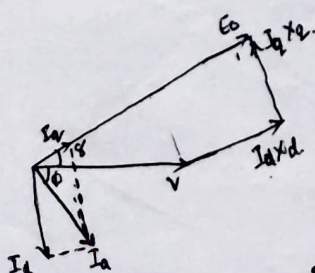


$\rightarrow$ Reduced vol. apply $\rightarrow$ Armature.

$\rightarrow$ Rotor rotates $<$ syn speed.

$$x_q = \frac{\text{min. vol}}{\text{max. current}}$$

$$x_d = \frac{\text{max. vol}}{\text{min. current}}$$



$$I_q = I_a \cos(\delta + \phi)$$

$$I_d = I_a \sin(\delta + \phi)$$

$$E_o = V \cos \delta + I_d x_d$$

$$P_{out} = V I_a \cos \phi$$

$$P_{in} = V [I_a \cos(\delta + \phi) + I_d \sin(\delta + \phi)]$$

- Advantages

1. Continuity of service.

- If one alternator fails, the continuity of supply can be maintained through the other healthy units.

- ensure uninterrupted supply

2. Efficiency

- Alternators give max efficiency when delivering F.L. o/p.

- we can turn on or off according to the requirement of load.

3. maintenance & repair:

- During maintenance & repair of one alternator the other can continue supply.

4. Load growth

- load demand rises day by day, so we can met by adding alternator parallel. 1

5. Cost of additional $\downarrow$.

Conditions.

1. Terminal voltage = bus bar vol.
2. frequency of generated vol = bus bar freq.
3. phase voltage = phase vol of bus bar.
4. phase sequence (R Y B) = bus bar sequence.

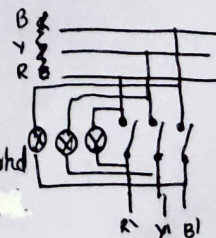
methods of synchronisation

1. Dark lamp method.

- PM of incoming alternator started and brought up close to track

- Alternator is then excited if its voltage is brought up to the bus bar vol. by varying field excitation

- lamps are watched closely



connected, then 3 lamps

will become bright & dim together

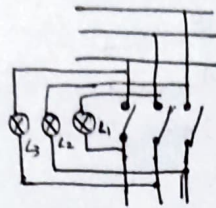
- If they bright & dim in sequence, then the phase sequence of alternator must be reversed.

- The speed PM is adjusted, so that the lamps flicker at very low rate.

- Then vol. of alternator again adjusted & the paralleling switch is closed when all 3 lamps are dark.

- Bright lamp method.

- $L_1 \& L_3 \rightarrow$ Bright
 $L_2 \rightarrow$ Dark.



- 3 lamps are placed in ring, a light wave travelling in clock wise $\rightarrow$ fast

anti clockwise $\rightarrow$ slow

- paralleling switch is closed when the change in light are slow & when L_2 is dark.

- Synchroscope.

- If phase sequence is known
Then the best method of synchronization is synchroscope.



Voltage regulation.

$$E_0 = \sqrt{(V \cos \phi + I_a R_a)^2 + (V \sin \phi \pm I_a X_s)^2}$$

$+$ $\rightarrow$ lag
 $-$ $\rightarrow$ lead.

$$E_{mf} \text{ at FL} = V_0 + I_a R_a \cos \phi$$

$I_{f1} \rightarrow$ voltage
 $I_{f2} \rightarrow$ current

Emf method

$$I_f = \sqrt{(I_{f1})^2 + (I_{f2})^2}$$

$I_{f1} \rightarrow$ correspond to E_0
 $I_{f2} \rightarrow$ correspond to Emf E .

Then note Emf value correspond to I_f from OCC.

Lead or lag.

$$I_f = \sqrt{I_{f1}^2 + I_{f2}^2 + 2 \times I_{f1} \cdot I_{f2} \cdot \cos(180 - (90 \pm \phi))}$$

$+$ $\rightarrow$ lag
 $-$ $\rightarrow$ lead.

find E_0 with I_f from graph.

ZPF.

• more accurate

$E \rightarrow$ value of V_L from graph.

$I_{f1} \rightarrow$ correspond to E .

$I_{f2} \rightarrow$ EB $\rightarrow$ armature reaction current

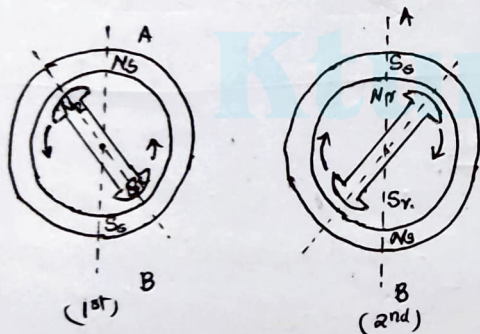
$I_f \rightarrow$ calculate.

find E_0 from OCC with I_f .

Synchronous motor - Principle.

- 2 main parts - stator & rotor.
- 3 ϕ supply $\rightarrow$ stator $\rightarrow$ produce rotating mag field at syn. speed.
- DC supply $\rightarrow$ rotor $\rightarrow$ strong electromagnet.
- If a electromagnet is present in a rotating mag. field, the electromagnet is magnetically locked with the rotating magnetic field and rotates with same speed of rotating field.
- speed of rotor = speed of rotating mag field.
ie, $N_s = \frac{120f}{P} \rightarrow$ only depend on f & P .

Why syn. motor is not self starting



$\rightarrow$ (1st) - fig.

N_s & N_r will repel each other and experiences a torque in anticlockwise direction

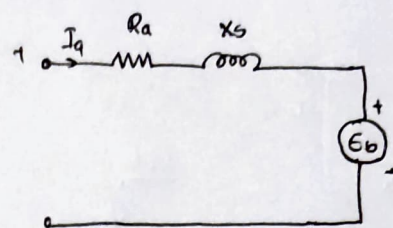
(2nd)

After a period of half cycle, stator poles changes their position but rotor poles remains at previous position

$\rightarrow$ (2nd) - fig

Now S_s & N_r attract each other & rotor experiences a torque in clockwise direction
- so torque on rotor is rapidly reversing it is not unidirectional. Due to large inertia of rotor, rotor does not respond to changing torque.

Equivalent ckt.



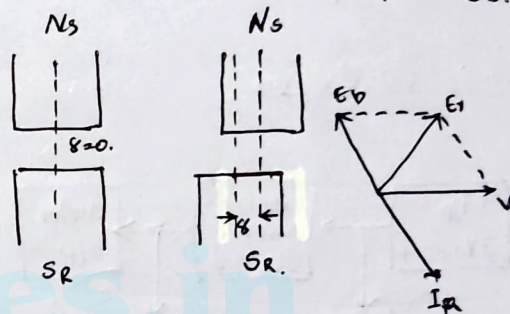
$$E_b \propto \text{flux}$$

$$\text{flux} \propto I_f$$

$$\therefore E_b \propto I_f$$

Syn motor on load.

$$E_r = V - E_b$$



- If motor is on load, the rotor poles tend to fall back by an angle θ .
- As the load $\uparrow$ the resultant voltage E_r $\uparrow$
- Thus motor draws large current

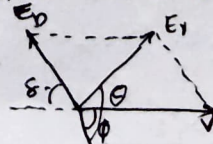
$$I_a = \frac{V - E_b}{Z_s}$$

Pull out torque.

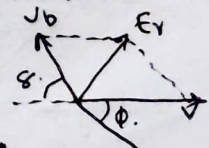
The load torque at which the motor pulls out of synchronism is called pull out torque.

Phasor diagram

Motor on No-LOAD.



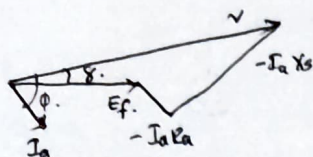
Motor on LOAD



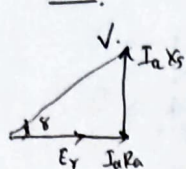
$$\theta \uparrow \text{ as } I_a \uparrow$$

Phasor diagram

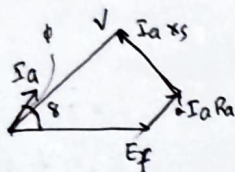
• Lagging p.f.



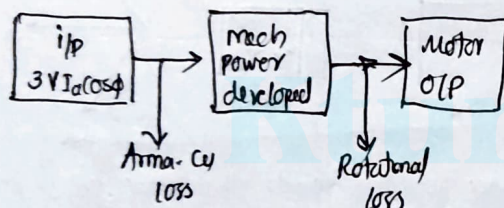
UPF



• Leading p.f.



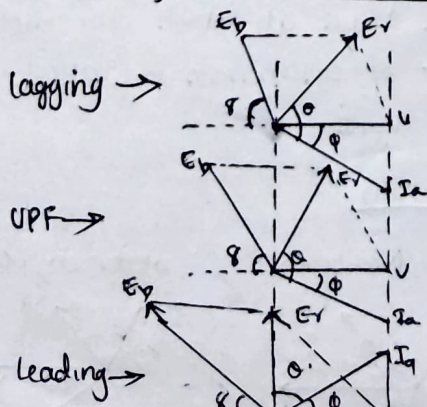
Power flow diagram



Losses

- Load losses - Arms. Cu loss.
- OC losses - Frictional loss, Windage, Iron loss. (hysteresis, Eddy current)

Effect of varying excitation on Ia & p.f.



V-curve

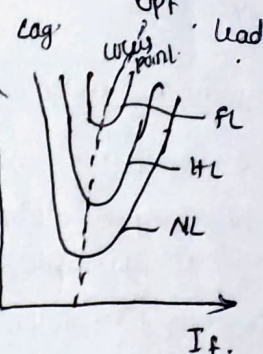
graph plotted b/w Ia & If

Excitation of motor changed, the magnitude of Ia changes.

Ia is min → UPF

→ increases → from UPF to lead or lag

→ since it is V shaped it is called V-curve.



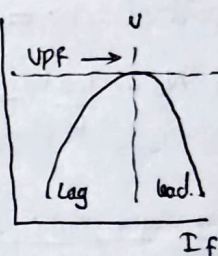
Inverted V-curve

graph b/w If & p.f.

Excitation changes, p.f. also changes.

Under excited → Lag

Over excited → lead.



3φ IM.

- Stator
- Rotor { squirrel cage rotor, wound rotor.

$$S = \frac{N_s - N}{N_s} \times 100$$

$$\frac{1700 - 1600}{1700} \times 100 = 0.0588 \approx 0.06$$

$$f_r = S \cdot f$$

$$\text{Rotor induced EMF / phase} = S \cdot E_2$$

$$\text{Rotor frequency } f_r = S \cdot f$$

$$\text{Rotor reactance / phase } X_r = S \cdot X_2$$

$$\text{Rotor current / phase } I_2 = \frac{S \cdot E_2}{Z_2} = \frac{S \cdot E_2}{\sqrt{R_2^2 + (S X_2)^2}}$$

$$\text{Rotor power factor } \cos \phi_2 = \frac{R_2}{Z_2} = \frac{R_2}{\sqrt{R_2^2 + (S X_2)^2}}$$

→ Electrical equivalent of mech
 Power developed in rotor = $R_2 \cdot \frac{1-s}{s}$

→ Total electrical power i/p to rotor

$$P_g = \text{Rotor cu loss} + \text{Mech power in rotor}$$

$$= s \cdot P_g + R_2 \cdot \frac{1-s}{s}$$

Torque.

$$P_{\text{mech}} = R_2 \cdot (1-s)$$

1. under running condition $R_{\text{cu loss}} = \frac{P_2 \cdot s}{\text{air gap}}$

$$T = \frac{k \cdot s \cdot R_2 \cdot E_2^2}{R_2^2 + (s \cdot X_2)^2}$$

2. starting torque

$$T_{\text{st}} = \frac{k_1 \cdot R_2}{R_2^2 + X_2^2} \quad s = 1$$

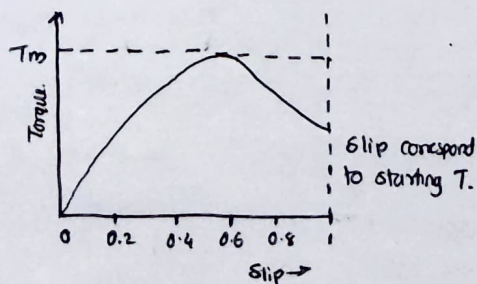
3. Full load torque.

$$T_f = \frac{k \cdot s_f \cdot R_2 \cdot E_2^2}{R_2^2 + (s_f \cdot X_2)^2} \quad s_f \rightarrow \text{full load slip.}$$

4. Pull out / maximum / Breakdown torque.

$$T_{\text{max}} = \frac{k \cdot E_2^2}{2 X_2}$$

Torque-slip chara.



$$T = \frac{k \cdot s \cdot R_2 \cdot E_2^2}{R_2^2 + (s \cdot X_2)^2}$$

→ When speed = N_s & slip = 0 $\Rightarrow T = 0$.

→ speed near to syn speed, slip is very small
 value of $s \cdot X_2$ is very small compared to R_2 .

i.e. $T \propto s$ (since R_2 is constant)

- As slip increases, speed $\downarrow$ & $T \rightarrow \uparrow$

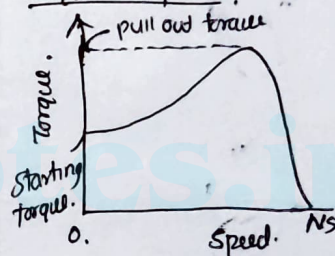
T reaches max $\rightarrow s = \frac{R_2}{X_2}$

- with further $\uparrow$ of $s \rightarrow T \downarrow \rightarrow$ motor slows down - stops.

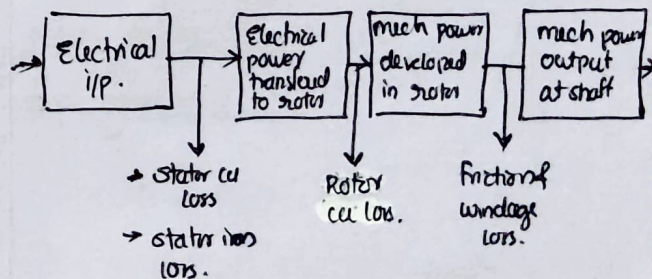
- at higher value of s , R_2 is negligible to $s \cdot X_2$.

$T \propto \frac{s}{(s \cdot X_2)^2} \Rightarrow T \propto \frac{1}{s} \Rightarrow$ Curve is hyperbola.

Torque-speed.



Power stage



1. fixed loss.

- stator iron loss
- Friction & windage loss.

2. Variable loss.

- stator cu loss
- Rotor cu loss.

Equivalent ckt.

Equations → Syn motor

$$P_{in} = \frac{P_{out}}{\eta}$$

$$I_p = \frac{P_{in}}{\sqrt{3} V_L \cos \phi} \quad E_r = I \cdot Z_s$$

$$|Z_s| = \sqrt{R^2 + j\omega L^2}$$

$$\text{Internal angle } \phi = \tan^{-1} \frac{X_s}{R_a} \quad \begin{array}{l} + \text{ - leading} \\ - \text{ - lag} \end{array}$$

Induced emf

$$E_{phn} = \sqrt{V_p^2 + E_v^2 + 2V_p E_v \cos(\phi + \phi)}$$

mech power

$$P_{mech} = P_{in} - \text{cu loss} \quad \leftarrow 3 \times I_a^2 R_a$$

$$P_{out} = \frac{2\pi N_s T}{60} \Rightarrow T = \frac{60 \times P_{out}}{2\pi N_s T}$$

Ktunotes.in

- Clogging.

- At certain times, even when full voltage is applied to the stator winding, the squirrel cage IM refuses to start & pick up speed.

- Due to magnetic locking b/w stator and rotor teeth.

- Occur when no. of stator slots is the integral multiple of Rotor slots.

- mainly occur in squirrel cage IM.

- slip-ring have high starting torque.

- can be eliminate by making $S_1 \neq S_2$ and skewing the rotor.

- Crawling.

- 3 ϕ IM, especially squirrel cage IM exhibit a tendency to run stably at a sub synchronous speed ($\frac{1}{7}$ th of syn speed) & unable to pick up normal speed.

- If the vol. applied to the stator is non sinusoidal or contain harmonics then the air gap flux won't be sinusoidal.

- Each harmonic voltage produce its own flux & torque $\rightarrow$ causing noise & vibration

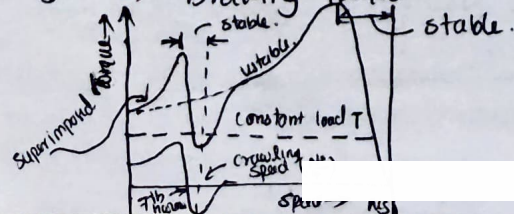
- The major harmonics $\rightarrow$ 3rd, 5th, 7th

- 3rd harmonics are absent. $\rightarrow$ balanced 3 ϕ load.

- total flux contain $\rightarrow \phi_1 - \phi_5 + \phi_7$.

- 1st & 7th $\rightarrow$ motoring torque

- 5th $\rightarrow$ braking torque.



- $S_1 \neq S_2$.

- Difference b/w S_1 & $S_2 \Rightarrow \pm P, \pm 2P$ or $\pm 3P$

- Difference b/w S_1 & S_2 not be $\Rightarrow P \pm 1$ or $P \pm 2$

- Reduce 5th & 7th harmonic by short pitch winding

Double cage IM.

- Stator is same as SCIM.

- Rotor carry 2 set of SC winding

- Inner cage - low resistance + high reactance

- Outer cage - High resistance + low reactance
Manganese Brass.

- Inner cage has higher cross-sectional area than outer cage.

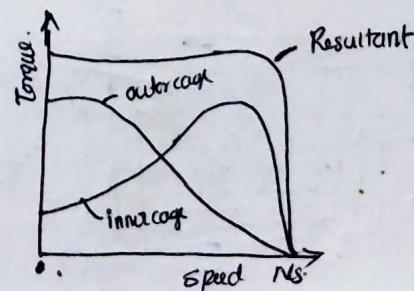
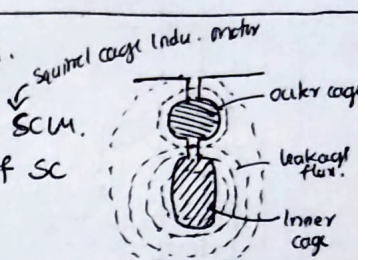
- During starting, $R_r = R + \frac{1}{2}$ of inner cage winding is high reactance is low so.

- Impedance of outer cage is low & most of the current flows through the outer cage high resistance winding.

$\therefore$ high starting torque is developed.

- During normal speed, slip $\downarrow$ & reactance of inner cage $\downarrow$ s.

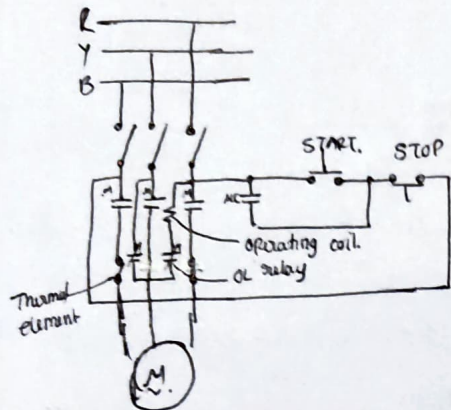
- Because of low resistance, inner cage carry more current & produce running torque.



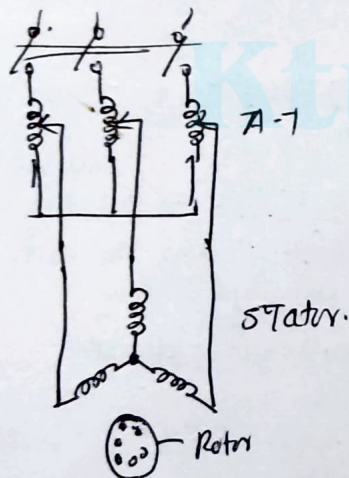
Starters.

- limit starting current
- protection against overload & under voltage.
- SRM $\rightarrow$ Improve starting torque

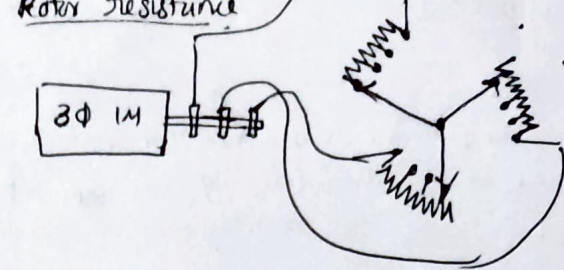
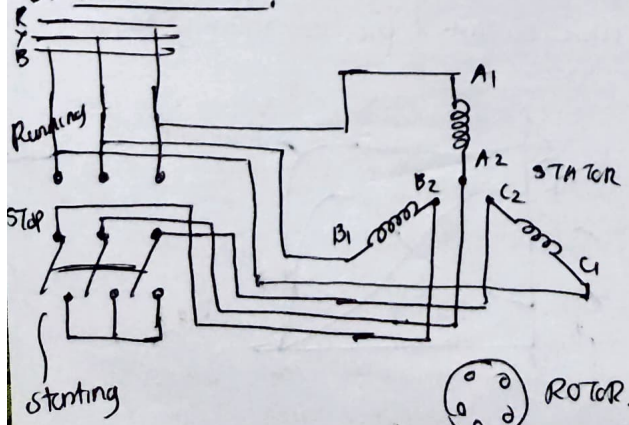
DOL.



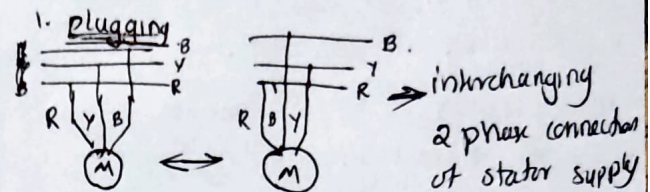
2. Autotransformer.



3. Star-Delta.



Braking of IM.



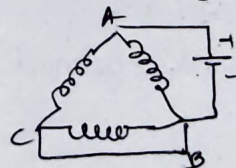
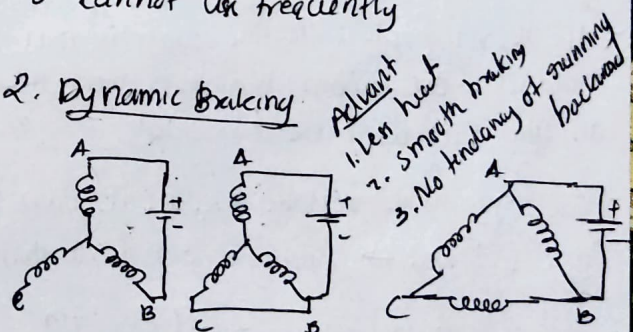
Advantages

1. quickest method.
2. cheaper method

Disadvantages

1. power wasted as heat
2. heating up of machine
3. cannot use frequently

2. Dynamic Braking



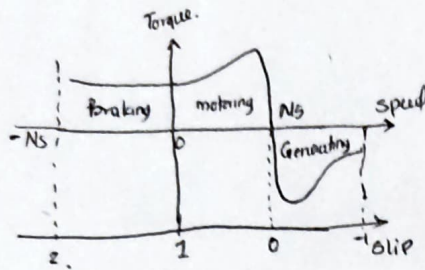
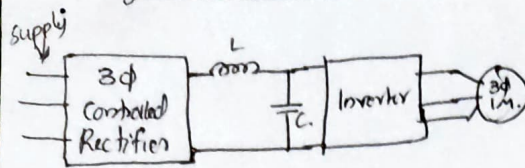
$\rightarrow$ Stator windings are disconnected from supply.

$\rightarrow$ Then connect to a DC source as shown in the fig.

$\rightarrow$ When connecting to DC supply an magnetic field is set up in stator and act as a field excitation. The rotor rotating as armature of DC motor.

$\rightarrow$ This will set up a current in rotor and develop motoring torque. — Direction is opposite to the regenerative braking action $\rightarrow$ Set necessary

3. Regenerative braking.



- IM → generator.
- electrical power fed back to supply.
- generating action produces required braking torque.
- when speed of motor is fed above syn speed, then it acts as a induc. generator.
- Done by switching over to low frequency supply → frequency controlled electric drives
- During braking slip & torque is negative.
- Act as generator receiving mech energy and supply the power

Advans

- generated power is fully utilized.

Disad

- low variable frequency supply is needed.

Speed control

1. SCIM - SRIM

- varying stator supply vol.
- varying stator supply frequency
- varying stator no. of poles

2. SRIM

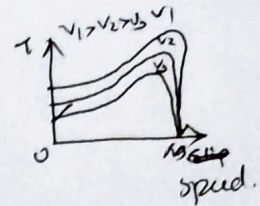
- Rotor resistance method
- cascade operation
- injecting external

$$N = \frac{120F}{P} (1-s)$$

1. stator vol. control

$$T \propto E_2^2 \Rightarrow E \propto V$$

$$\therefore T \propto V^2$$



by varying T , we can change s .

from T -slip chara it is clear.

- variable vol. const frequency supply

Advanta

- Simple
- low cost

Disad

- vol. cont ↑ beyond rated value. speed control below rated vol is possible.
- large change in voltage is required to get a small speed change.
- Torque developed is changed

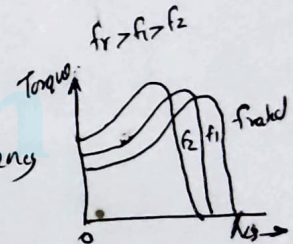
2. V/f control

- varying supply frequency

$$\Phi \propto \frac{V}{f}$$

- $f \downarrow$ to reduce speed, flux $\uparrow$ & causes saturation of core.

- $f \uparrow$ to $\uparrow$ speed, flux $\downarrow$ & torque $\downarrow$
we have to maintain $\frac{V}{f}$ a ratio constant



Advans

- smooth speed control
- above & below rated speed is possible.

Disad

- V/f is required.
- costly & complicated

3. Rotor resistance method

Adv-

- simple, low cost
- starting torque

Disad

- losses & efficiency $\downarrow$
- speed $\rightarrow R_2$ & load.